

FINGAL, Australia

WMO: 959740

Lat: 41.6428S

Lon: 147.9664E

Elev: 237

StdP: 98.51

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -3.3	(c) -2.3	(d) -4.7	(e) 2.6	(f) 0.4	(g) -3.8	(h) 2.8	(i) 1.5	(j) 10.3	(k) 11.2	(l) 8.6	(m) 10.2	(n) 0.6	(o) 140	(p) 0.532

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 12.5	(c) 28.7	(d) 16.9	(e) 26.5	(f) 16.6	(g) 24.4	(h) 15.9	(i) 19.2	(j) 24.4	(k) 18.2	(l) 23.3	(m) 17.2	(n) 22.0	(o) 5.1	(p) 30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	12.9	20.8	16.4	12.0	19.8	15.4	11.3	19.1	55.6	24.5	52.3	23.2	49.4	21.8	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.4	7.2	6.4		-5.4	34.0	1.2	3.0	-6.2	36.2	-6.9	37.9	-7.6	39.6	-8.4	41.8	
				DB												
				WB												
					-5.7	21.3	1.2	0.9	-6.6	22.0	-7.3	22.5	-8.0	23.0	-8.8	23.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	16.8	16.1	14.6	11.5	8.9	7.0	6.6	7.2	8.9	10.8	13.1	14.9
	DBStd	4.48	3.23	2.77	3.08	2.78	2.64	2.42	2.20	2.28	2.52	2.98	3.06	2.99
	HDD10.0	443	0	0	1	14	53	94	109	91	50	24	5	1
	HDD18.3	2602	69	74	122	204	293	339	365	344	282	234	161	114
	CDD10.0	933	211	170	145	61	19	4	2	5	18	49	97	152
	CDD18.3	51	21	10	7	1	0	0	0	0	0	1	3	8
	CDH23.3	701	304	99	81	7	0	0	0	0	0	21	68	122
	CDH26.7	198	109	19	18	1	0	0	0	0	0	2	17	33
Wind	WSAvg	2.6	3.2	2.8	2.5	2.2	2.4	2.1	2.3	2.6	2.7	2.9	3.0	3.1
Precipitation	PrecAvg	587	48	35	43	51	50	50	55	64	49	47	49	55
	PrecMax	957	192	197	195	199	246	195	171	190	139	147	149	201
	PrecMin	374	2	0	3	0	3	14	6	18	4	5	11	5
	PrecStd	143	39	37	36	48	50	37	35	40	31	28	28	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	29.4	28.8	24.0	19.2	15.7	15.0	17.0	20.7	26.2	29.1	30.5
		MCWB	18.6	17.3	17.1	15.2	12.9	11.4	9.9	10.5	12.3	13.7	16.2	17.8
	2%	DB	29.3	26.1	25.4	21.5	17.3	14.5	13.7	15.2	18.1	22.0	25.1	26.7
		MCWB	17.4	17.6	16.6	14.2	11.9	10.6	9.4	9.8	11.1	12.4	15.1	16.0
	5%	DB	26.5	24.1	22.9	19.3	15.9	13.4	12.7	13.8	16.3	19.4	22.1	23.9
		MCWB	16.7	16.6	15.9	13.4	11.2	9.8	8.9	9.3	10.4	12.4	14.6	15.5
	10%	DB	23.9	22.1	20.8	17.5	14.7	12.3	11.7	12.7	14.8	17.4	19.9	21.7
		MCWB	15.8	15.7	15.0	12.7	10.7	9.2	8.3	8.7	9.8	11.5	13.6	14.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.0	20.0	19.2	16.9	14.2	13.5	11.5	11.9	14.1	16.0	18.5	19.2
		MCDB	28.7	24.5	24.8	20.7	15.9	14.5	13.3	14.9	18.4	19.2	23.2	26.2
	2%	WB	19.4	18.9	18.1	15.5	12.9	11.4	10.1	10.8	12.3	14.4	16.6	17.8
		MCDB	24.7	23.2	22.5	19.1	16.1	13.2	12.6	13.8	16.0	18.6	22.7	23.2
	5%	WB	18.1	17.8	16.9	14.5	12.1	10.3	9.4	9.9	11.3	13.3	15.5	16.6
		MCDB	23.8	22.1	21.0	18.1	15.1	12.4	12.0	13.0	15.0	18.1	20.8	21.5
	10%	WB	16.9	16.8	15.8	13.3	11.1	9.5	8.6	9.1	10.4	12.1	14.4	15.4
		MCDB	22.5	20.7	20.1	16.9	14.3	11.9	11.3	12.2	14.1	16.7	19.2	20.5
Mean Daily Temperature Range	5% DB	MDBR	12.5	11.0	11.8	11.8	10.9	10.1	10.2	10.7	11.5	11.8	11.7	11.9
		MCDBR	17.8	14.2	15.2	15.1	12.8	11.4	11.2	12.6	13.9	15.3	16.1	16.4
		MCWBR	7.5	7.1	7.5	8.7	8.4	8.1	7.8	8.2	8.2	8.3	8.0	7.8
	5% WB	MCDBR	14.0	11.7	12.3	12.4	10.7	9.7	10.0	10.7	11.9	12.7	13.6	13.3
		MCWBR	7.1	6.7	6.6	7.7	7.5	7.4	7.3	7.4	7.7	7.7	7.7	7.2
Clear-Sky Solar Irradiance	taub		0.335	0.329	0.320	0.310	0.294	0.288	0.285	0.300	0.315	0.323	0.329	0.331
	taud		2.540	2.575	2.594	2.596	2.611	2.614	2.632	2.587	2.537	2.525	2.535	2.540
	Ebn at Noon		992	972	938	880	828	799	831	878	928	967	992	1002
	Edh at Noon		107	98	88	75	63	57	60	74	91	102	107	108
All-Sky Solar Radiation	RadAvg		6.74	5.66	4.47	3.04	2.00	1.57	1.78	2.60	3.86	5.17	6.07	6.83
	RadStd		0.47	0.27	0.21	0.17	0.11	0.12	0.08	0.19	0.26	0.26	0.44	0.42

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	DBAvg	+0.42	N/A	-0.84	+1.41	N/A	N/A	N/A	+126	+20
(47) Regional (27 neighbors)	DBAvg	+0.33	N/A	N/A	+0.70	+0.32	N/A	N/A	-102	+109

Nomenclature: See separate page